

Term Project: LSTM vs Transformer Encoder on AG News Classification

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AI Usage Appendix

Tools Used

Codex was used for project planning, code drafting support, debugging and document review.

What AI Was Used For

AI was used to brainstorm a controlled experiment plan, check the scaffold against dataset-source and test-set rules, and review PyTorch notebook components. For code-level work, AI assistance was used to review the structure. All generated code was inspected and run cell-by-cell, and the masked mean pooling and Transformer padding-mask logic were manually checked against the attention masks.

Incorrect, Incomplete, Or Misleading AI Outputs

During code review, no instances were found of common AI-assisted code errors: applying ``softmax`` before ``CrossEntropyLoss``, normalizing confusion matrices when raw error counts were needed, or using the official test split during model selection or ablation.

Team Modifications And Decisions

The team kept the project constrained to one dataset source, Hugging Face ``fancyzhx/ag_news``, and required the official test split to be used only for final evaluation. The team selected dataset-size ablation as the first ablation and kept sequence length as the second option. We also added explicit checks for integer labels, train-only vocabulary construction, padding masks, and reproducible seeds rather than relying on generated code without verification.

Effect Of AI Use

AI reduced the time needed to organize requirements, audit deliverables, draft code scaffolds, and write clear explanations.